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EmoBipedNav: Emotion-Aware Social Navigation for Bipedal Robots With Deep Reinforcement Learning

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***Abstract*—This study presents an emotion-aware navigation framework—EmoBipedNav—using deep reinforcement learning (DRL) for bipedal robots walking in socially interactive environments. The inherent complex dynamics of bipedal robots challenge their safe maneuvering capabilities in dynamic environments. Furthermore, the intricacies of social interactions and cues such as emotions significantly compound the challenges of bipedal robot navigation. To address these coupled issues, we propose a two-stage pipeline that jointly considers the bipedal full-body dynamics and the complexities of socially aware navigation. More specifically, an emotion-integrated navigation policy is developed to balance safety, efficiency, and social courtesy by responding to human emotional cues. One key component of the policy is a novel representation of social environments using sequential LiDAR grid maps, from which we extract latent features, implicitly including collision regions, discomfort zones determined by emotions, social interactions, and the evolving dynamics of the navigation system including robot movements and pedestrian motions. To accounts for path tracking errors and locomotion constraints during social navigation, we present an end-to-end navigation system that incorporates full-order robot dynamics during training. Finally, extensive benchmarking and sim-to-real experiments demonstrate that our method outperforms both model-based planners and DRL baselines, and generalizes effectively to real-world environments.**

***Index Terms*—Bipedal robot, deep reinforcement learning (DRL), emotion-aware navigation, locomotion, social navigation.**

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The hardware videos and open-source code are available at https://gatech-lidar.github.io/emobipednav.github.io/.

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I. Introduction

THE research and development of bipedal robots have attracted considerable attention in the robotics community, largely due to their human-like morphology and versatile locomotion capabilities [1], [2], [3], [4], [5], [6]. While substantial progress has been made in developing stable locomotion controllers, a long-term objective is to empower bipedal robots to navigate autonomously and courteously in pedestrian-rich environments with social awareness. However, achieving safe and socially compliant navigation for bipedal robots remains a significant challenge. This difficulty stems from the complex dynamics of bipedal locomotion, limited maneuverability in dynamic settings, and the need to appropriately interpret and respond to pedestrian interactions and social cues such as emotions.

A key challenge in social navigation lies in modeling implicit interactions among agents, including human–human and human–robot dynamics. Conventional industry solutions, particularly in autonomous driving, often neglect these interactions by independently predicting each pedestrian's trajectory without considering the influence of other pedestrians [7]. Recent end-to-end approaches have shown promise in capturing implicit interactions using attention mechanisms, graph neural networks, recurrent neural networks, and so forth [8], [9]. However, these methods typically assume fully observable environments with access to detailed pedestrian states such as positions, velocities, and preferred speeds—assumptions that demand sophisticated and often impractical perception systems for real-world deployment. An alternative direction involves directly mapping raw sensor data to robot actions [10]. While this simplifies perception, such methods often require hybrid policies to guarantee safety, partially depend on pedestrian state estimation, or suffer from high training overhead.

In addition to social interactions, social cues—such as human emotions—psychologically influence preferred personal space

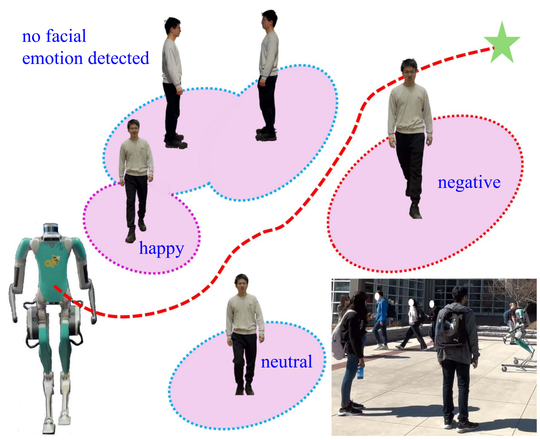


Fig. 1. Emotion-aware social navigation with the bipedal robot Digit.

during interactions with other agents, including both pedestrians and robots [11], [12], as illustrated in Fig. 1. Despite their importance, emotion-aware social navigation remains largely underexplored [13], [14]. Moreover, existing studies assume the presence of only a single pedestrian in the environment, significantly simplifying the complexity of real-world social navigation scenarios.

Although significant efforts have been dedicated to the use of wheeled mobile robots for social navigation, few studies address bipedal social navigation. Cutting-edge bipedal navigation studies in crowded environments decouple navigation and locomotion, planning navigation trajectories with reduced-order models (ROMs) at first, and then tracking the trajectories with locomotion controllers based on full-order dynamics [15], [16]. Such a planning architecture cannot take into account the tracking errors and constraints of low-level controllers, such as joint torque limits, for bipedal navigation. In addition, pedestrian trajectory prediction is also individually executed [16] because prediction models are pretrained using specific datasets only including pedestrian–pedestrian interactions, ignoring pedestrian–robot interactions. Such pretrained models might further cause deployment discrepancies in bipedal robots. To achieve socially aware bipedal navigation, a unified policy is needed that enables interactive navigation while simultaneously accounting for bipedal locomotion. Deep reinforcement learning (DRL), as an end-to-end framework, offers a promising solution by integrating navigation and locomotion within a single architecture. It enables autonomous and socially courteous navigation in pedestrian-rich environments while considering full-body bipedal dynamics.

In this study, we present an emotion-aware navigation framework— EmoBipedNav—using DRL for bipedal robots walking in socially interactive environments. We represent the environment using sequential LiDAR grid maps (LGMs), from which we extract latent features, implicitly including collision regions, discomfort zones, social interactions, and the evolving dynamics of the navigation system. We employ a DRL-based navigation architecture to map these latent features directly to bipedal robot actions, specifically, the sagittal center-of-mass (CoM) speed derived from ROMs and the heading angle of Digit's torso. To address the discrepancies between ROMs and full-order dynamics, we train the social navigation policy directly in a physics-based simulator with a full-body locomotion controller. In addition, we benchmark against state-of-the-art (SOTA) navigation pipelines to highlight the efficacy of our framework. The main contributions are summarized as:

1) *Novel DRL observation representation:* We introduce LGMs as a novel representation of highly dynamic environments integrated with pedestrian emotions.
2) *Emotion integration in socially interactive environments:* Our framework incorporates pedestrian emotions and constructs various discomfort zones, enabling emotion-aware navigation in socially interactive environments.
3) *End-to-end navigation framework:* We propose an end-to-end navigation pipeline that directly maps latent features to bipedal robot actions and is trained with full-body bipedal robot dynamics, effectively mitigating model discrepancies induced by ROMs.
4) *Benchmarks and hardware evaluations:* We compare the performance of our method with SOTA baselines. In addition, we transfer the simulated navigation policy to the real bipedal robot Digit.

II. Related Work

Navigation with bipedal robots: Numerous studies have investigated bipedal robot navigation in relatively simple environments with static and sparsely distributed obstacles [4], [17], [18], [19]. Recent approaches have begun to address dynamic environments by incorporating moving obstacles alongside static barriers [5], [20], [21], [22]. However, these dynamic obstacles are sparsely distributed around the robot, e.g. only one single moving obstacle is in close proximity to the ego-agent robot. Moreover, the interactions are oversimplified, assuming slower moving obstacles and neglecting social interactions between pedestrians and robots [21].

Research specifically addressing crowd navigation for bipedal robots remains largely underexplored, with the exception of the recent study in [16], which assumes that bipedal robots and pedestrians adhere to the same social rules, that is, robots learn their navigation path from real pedestrian datasets and imitate pedestrian navigation strategies. However, this assumption is restrictive, as bipedal robots exhibit fundamentally different dynamics and face more stringent physical constraints compared to pedestrians. Furthermore, current collision avoidance strategies are typically conservative because robots risk freezing due to the stochastic and unpredictable nature of pedestrian trajectory predictions [23].

Social navigation with DRL: Although research on social navigation with bipedal robots remains limited, there exists a substantial body of work focused on social navigation for wheeled mobile robots [8]. In particular, end-to-end frameworks employing DRL have been widely adopted to map environmental states directly to robot actions [9]. A notable example is the collision avoidance with deep reinforcement learning (CADRL) approach [23], which formulates the navigation problem as a Markov Decision Process (MDP) under the assumption of fully observable environments, requiring pedestrian attributes such as size, position, velocity, and preferred speed. However, CADRL handles interactions by pairing the ego-agent with each pedestrian individually, thereby ignoring the influence of pedestrian-to-pedestrian interactions.

To address this, later studies such as Social Attention RL (SARL) [24] and Relational Graph Learning (RGL) [25] aim to model implicit social interactions among pedestrians to generate socially compliant navigation behaviors. Nonetheless, these SOTA methods still rely heavily on the assumption of full observability, which in practice demands complex perception and prediction systems that are difficult to implement in real-world scenarios. Furthermore, they presume that the robot's kinematics are similar to those of pedestrians and largely unconstrained, an assumption that does not hold for bipedal robots due to their restrictive locomotion capabilities.

While MDPs are designed for decision making in fully known environments, Partially Observable MDPs (POMDPs) are suitable for decision making in socially interactive environments, which involves partially observable pedestrian interactions and future trajectories [26]. SOTA approaches that map raw or preprocessed sensor data to robot actions can be formulated as POMDPs [27], [28], [29], [30]. An intuitive representation is the sequential LiDAR scans, which are stacked and encoded with convolutional neural networks (CNNs) [27] or directly processed with long-short-term memory (LSTM) [28]. Combining sequential LiDAR scans with fully known states (i.e., pedestrian position and speed), the DRL velocity obstacle (DRL–VO) method converts these observations and states into an image for feature extraction [29]. In practice, they are dependent on complex estimation techniques to obtain these states [29]. Moreover, their network architectures are susceptible to divergence, plateauing, and collapse when applied to robots with complicated dynamics and strict constraints [27], [28], [29]. Compared to others, our approach is more straightforward, with only sequential LiDAR scans converted into grid maps. In addition, the neural networks of our feature extractor followed by the soft actor-critic (SAC) DRL algorithm [31] yield a minimal risk of convergence failures for DRL training.

Emotion-aware navigation: Most of the studies discussed above simplify pedestrians as circles, overlooking their social cues such as emotions. Pedestrian emotions implicitly convey information, such as the preferred comfort distance that people prefer to maintain from others [11]. Psychological experiments [12] have identified comfort distances associated with various emotions. With these insights, social robots can adaptively maintain a customized comfort distance from pedestrians with specific emotions [14]. However, these emotion-aware robot navigation approaches focus primarily on the emotions of individuals, neglecting the complex interactions that occur among pedestrians with various emotional states, which will be addressed in our study.

III. Approach

In this section, we first introduce the reduced-order model (ROM) of bipedal locomotion and the partially observable Markov decision process (POMDP) of the social navigation problem. Subsequently, we present our DRL-based approach, detailing the novel observation space, the DRL network architecture, and the design of the integrated reward functions. The overall framework is illustrated in Fig. 2.

A. Preliminaries

The dynamics of bipedal robots is commonly approximated using ROMs, such as the linear inverted pendulum (LIP) model [32], [33], as depicted in Fig. 2(c). The discrete state transition equation for the LIP model at the k^{th} walking step in the local sagittal frame is given as follows:

$$d_k = \frac{\upsilon_k^{\text{loc}} \cosh(\omega T) - u_k^{\upsilon}}{\omega \sinh(\omega T)}$$
$$\Delta x^{\text{loc}}(d_k) = \upsilon_k^{\text{loc}} \frac{\sinh(\omega T)}{\omega} + d_k\left(1 - \cosh(\omega T)\right) \quad (1)$$

where the control input u_k^{υ} represents the desired sagittal velocity at the next foot placement switch instant, the output state $\Delta x^{\text{loc}}(d_k)$ is the local sagittal CoM position increment between two consecutive walking steps, υ_k^{loc} denotes the sagittal velocity in the local coordinate at the k^{th} walking step switch instant, d_k is an auxiliary variable that specifies the sagittal foot distance relative to CoM, T is the constant footstep time, the term $\omega = \sqrt{g/H}$ is derived from the gravitational constant g and the desired CoM height H. The ROM in (1) is equivalent to the formulation of the angular momentum LIP planner (ALIP) model in [32]. However, since our DRL network outputs the sagittal speed, our formulation uses the current sagittal velocity of CoM υ_k^{loc}, instead of the angular momentum about the contact point used in ALIP.

Since u_k^{υ} influences only the translational motion, we introduce a heading angle $u_k^{\Delta\theta}$ to steer the CoM orientation in the horizontal plane, as illustrated in Fig. 2(c). Combining the translational and rotational motions of the CoM, we define the action of our navigation policy as

$$\mathbf{u}_k = \{u_k^{\upsilon}, u_k^{\Delta\theta}\},\ 0 \leq u_k^{\upsilon} \leq \bar{u}^{\upsilon},\ \mid u_k^{\Delta\theta} \mid \leq \bar{u}^{\Delta\theta} \quad (2)$$

where $\bar{u}^{\upsilon}$ and $\bar{u}^{\Delta\theta}$ represent the maximum allowable sagittal speed and heading angle, respectively.

Given the control $\mathbf{u}_k$, the CoM displacement Δx^{loc}, and the LIP model, the CoM position is propagated from p_k to p_{k+1}, and the sagittal speed changes to $\upsilon_{k+1}^{\text{loc}}$ from υ_k^{loc}. However, there are discrepancies between the actual CoM position p_{k+1}^* and p_{k+1}, and between the actual sagittal speed $\upsilon_{k+1}^{\text{loc}*}$ and $\upsilon_{k+1}^{\text{loc}}$. More specifically, the full-body controller generates joint torques to track the desired CoM displacement Δx^{loc} and the control $\mathbf{u}_k$, as illustrated in Fig. 2(d). However, discrepancies between the LIP model and the full-body dynamics can cause the robot's states to deviate from the desired ones as experimentally illustrated in Section IV-C, which may lead to collisions and intrusions into discomfort areas of pedestrians, as demonstrated in Section IV-E.

Remark 1: In our previous work [16], the ROM in conjunction with a model predictive controller is used to plan feasible CoM trajectories for Digit. However, in this work, the ROM state [i.e.,

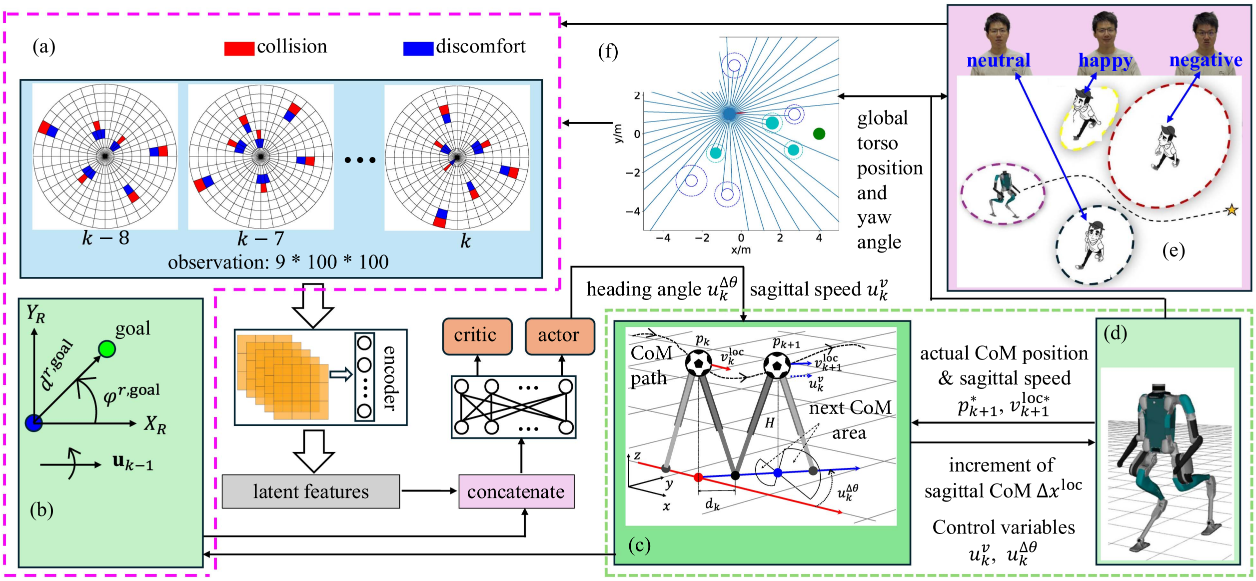


Fig. 2. Overview of the proposed EmoBipedNav framework using the bipedal robot Digit. (a) LiDAR grid maps where the red cells denote collision areas and the blue patches highlight discomfort zones. (b) States of goal position and last action. (c) LIP planner to regulate DRL actions. (d) Full-body controller to track CoM trajectory. (e) Preknown pedestrian emotions. (f) Raw LiDAR scans.

the heading angle and sagittal speed in (1)] is used to define the DRL actions. The input of the proposed network comes from the full-body simulation of Digit [see Fig. 2(d)], eliminating any discrepancies between the ROM and full-body dynamics, which exist in our previous work. Furthermore, we utilize the ROM for benchmarks, including both model-based motion planners and DRL-based baselines, effectively demonstrating the practical impact of robot dynamics discrepancies.

B. Problem Formulation

We formulate the social navigation problem as a POMDP, as the state information in socially interactive environments is inherently incomplete. Factors such as implicit social interactions and future pedestrian trajectories are not directly observable. To represent the social environment, we employ sequential LiDAR grid maps (LGMs). Although LGMs contain only partial information, they enable the extraction of rich implicit features, including collision zones, pedestrian–pedestrian and pedestrian–robot interactions, emotion-related discomfort areas, and the spatio-temporal dynamics of the navigation system.

In addition to LGMs which only contain partial information, we have two other fully observable states. For one thing, the goal position state is responsible for the target-directed navigation task. In addition, since pedestrians interact with the robot and the robot's actions influence pedestrian decisions, we also include the robot's actions as part of the states.

Given the observation of LGMs and the fully observable states, our objective is to optimize a navigation policy, straightforwardly projecting the observation and states into the robot actions. The problem is formulated as a POMDP defined by a 6-tuple $(S, O, A, \Gamma, R, \gamma)$, where S is the fully observable state space including the goal position and the robot's actions, O represents the observation space consisting of sequential LGMs, A stands for the action space, Γ denotes the state transition model fully propagated with the bipedal robot Digit, R is the reward function with an instantaneous scalar reward r_k at the kth walking step and γ is the discount factor. To derive the optimal navigation policy $\pi(a \in A | s \in S, o \in O)$, our goal is to maximize the expected sum of discounted rewards over an infinite time horizon, as follows:

$$V = \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_k \right]. \tag{3}$$

We will leverage a CNN-based encoder followed by a soft actor-critic (SAC) DRL algorithm [31] to iteratively optimize the navigation policy π.

C. Observation and State Spaces

We propose a novel representation for the observation space, referred to as pie-shape LGMs [see Fig. 2(a)], to comprehensively encapsulate the features of all entities around the ego-agent, including static obstacles having variable sizes and dynamic pedestrians with different emotions. Sequential pie-shape LGMs are built on the basis of raw LiDAR scans. More specifically, a 2-D LiDAR scan with K beams is emitted from the ego-agent to detect collision and discomfort margins, as illustrated in Figs. 2(f) and 3(a).

Remark 2: During training, we assume pedestrian emotions are pre-known and randomly assign them to facilitate sim-to-real transfer. These emotions are used to define corresponding discomfort zones, which contribute to the construction of the LiDAR grid maps. In real-world implementations, pedestrian emotions are estimated from facial cues using a pretrained CNN

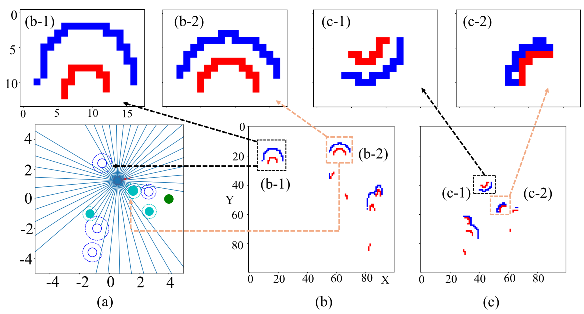


Fig. 3. LiDAR scan and its corresponding grid maps. (a) Raw LiDAR scan. (b) LiDAR grip map. (c) Occupation grid map. Top four figures detail the grids in (b) and (c).

model[1] consisting of six convolutional layers and four fully connected layers.

The detected obstacles are projected onto a pie-shape grid map [shown in Fig. 2(a)] with a maximum detection radius of L. This map is divided into M radial segments, forming M concentric rings. Each ring is further segmented angularly into M equal sections. The resolution of each grid increases as the ring gets closer to the center, ensuring that more details around the ego-agent are detected. Note that the pie-shape grid map is inherently circular and cannot be directly used for feature extraction with CNNs, which require a rectangular image input. To address this, the map is first flattened and restructured into a LiDAR grid map, as shown in Fig. 3(b).

Let $g(i, j)$ denote the grid cell in the ith ring and the jth angular segment. Each grid cell is intersected by K/M LiDAR beams passing through it. Consequently, we define the resolution of each grid cell as the number of beams per unit area, expressed as:

$$g_r(i, j) = \frac{KM^2}{\pi L^2 (2i - 1)} \tag{4}$$

which ensures that more detailed information, such as the collision cells and discomfort patches shown in Fig. 2(a), is captured for objects closer to the robot, allowing for more accurate local navigation around the ego-agent. Note that the pie-shape grid map is inherently circular and cannot be directly used for feature extraction with CNNs, which require a rectangular image input. To address this, the map is first flattened and restructured into a LiDAR grid map, as shown in Fig. 3(b).

Alternatively, occupancy grid maps (OGMs), [2] as shown in Fig. 3(c), are commonly used for local collision avoidance.[2] With the same map size M^2 and detection radius L, the resolution of each grid is calculated evenly as $(M/2L)^2$. In contrast, the resolution of our LGM is calculated on average as $(M/\sqrt{\pi}L)^2$, demonstrating that our LGM representation has a higher resolution on average. More specifically, the LGM has a higher resolution for grids closer to the map origin and a lower resolution for grids farther away. We set $M = 100$, $L = 6.0$ m, and $K = 1800$. For $i = 1$, the LGM has 100 grids while the OGM only has 4. When $i > 13$, the density of the OGM grid surpasses that of the LGM. As a result, the LGM can represent more details within the $13 \cdot L/M = 0.78$ m range of the ego-agent, as illustrated in the top figures of Fig. 3. Given the same obstacles close to the ego-agent, their representation in Fig. 3(b) (LGM) contains more pixels compared to Fig. 3(c) (OGM), indicating that LGM can capture more details when obstacles are near, which is beneficial for collision avoidance and alleviating discomfort, as demonstrated in Section IV.

In particular, we define a discomfort margin [blue grids in Figs. 2(a) and 3(b)] that expands according to the collision area [red cells in Figs. 2(a) and 3(b)]. The size of this discomfort zone is adjusted according to the emotional state of the pedestrian. We consider three emotional states: `happy`, `neutral`, and `negative` including angry, sad, fearful, and surprised. According to psychological research [12], happy emotions encourage approach behaviors and therefore shorter distances, whereas negative emotions promote avoidant behaviors and longer distances. Based on the studies in [12], [14], we assign the following distances from the robot margin to the pedestrian edge: 0.2 m for `happy`, 0.35 m for `neutral`, and 0.5 m for `negative` emotions.

The observation $\mathbf{o}_k$ at the current walking step is the LGM capturing only the static immediate environment in the spatial dimension, which might encourage conservative motions to avoid collisions as quantitatively analyzed in Section IV. To capture dynamic information in the temporal dimension, such as pedestrian-pedestrian and pedestrian-robot interactions, and the evolving dynamics of the navigation system, we incorporate previous LGMs from time step $k - N + 1$ to $k - 1$ as illustrated in Fig. 2(a). Thus, the observation in both the spatial and temporal dimensions is represented as $\mathbf{o}_k^{\text{sp-tem}} = \{\mathbf{o}_{k-N+1}, \ldots, \mathbf{o}_{k-1}, \mathbf{o}_k\}$.

In addition to the observation only including partial information, we define another two fully observable states as illustrated in Fig. 2(b): the position state of the goal $\mathbf{s}_k^{\text{goal}} = \{d^{r,\text{goal}}, \varphi^{r,\text{goal}}\}$, where $d^{r,\text{goal}}$ represents the distance between the goal and the robot and $\varphi^{r,\text{goal}}$ indicates the orientation of the goal in the robot frame; the last robot command $\mathbf{u}_{k-1}$. These two states are used for the goal-reaching task and the pedestrian–robot interaction, respectively.

D. Encoder-Integrated Networks

The encoder, as shown in Fig. 2, consists of CNNs followed by multilayer perceptrons (MLPs) to fully extract the latent features $\mathbf{f}_k^o$ from the sequential LGMs $\mathbf{o}_k^{\text{sp-tem}}$. Unlike SOTA approaches [24], [25] that rely solely on MLPs to process fully observable states, our encoder is capable of extracting more detailed and implicit information from observation LGMs. In addition, the proposed image-based observation integrates emotion-related discomfort zones, addressing a challenge faced by cutting-edge methods using raw sensor data [27]. Given these latent features $\mathbf{f}_k^o$, we concatenate them with the goal state $\mathbf{s}_k^{\text{goal}}$ and the action state $\mathbf{u}_{k-1}$. The combined inputs are fed into the SAC DRL framework [31] to optimize the navigation policy $\pi(\mathbf{u}_k | \mathbf{o}_k^{\text{sp-tem}}, \mathbf{s}_k^{\text{goal}}, \mathbf{u}_{k-1})$. The actor and value modules, as shown in Fig. 2, are made up of MLPs that take the same

[1]https://github.com/susantabiswas/realtime-facial-emotion-analyzer
[2]http://wiki.ros.org/costmap_ 2d

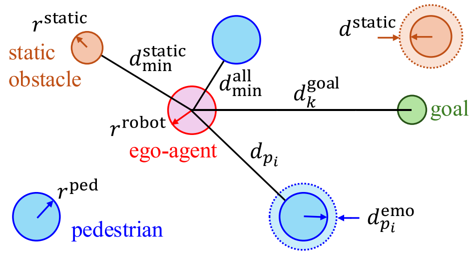


Fig. 4. Geometry representations of reward-related notations.

inputs $\{\mathbf{f}_k^o, \mathbf{s}_k^{\text{goal}}, \mathbf{u}_{k-1}\}$. In summary, the network structure can be expressed as follows:

$$\mathbf{f}_k^o = f_\vartheta(\mathbf{o}_k^{\text{sp-tem}}) \tag{5a}$$

$$v_k = f_\phi(\mathbf{f}_k^o, \mathbf{s}_k^{\text{goal}}, \mathbf{u}_{k-1}) \tag{5b}$$

$$\mathbf{u}_k \sim f_\theta(\mathbf{f}_k^o, \mathbf{s}_k^{\text{goal}}, \mathbf{u}_{k-1}) \tag{5c}$$

where the network parameters ϑ, ϕ, θ are updated using the SAC algorithm [31] and the exploration samples collected from the bipedal robot Digit, v_k represents the deterministic value function predicted by the critic network, and $\mathbf{u}_k$ is the action sampled from a Gaussian distribution with its mean and standard deviation parameters output from the actor network.

E. Reward Integration

The objectives of our navigation task are threefold: avoidance of collisions, reaching goals, and awareness of emotions. Accordingly, we define the reward function as follows:

$$r_k = r_k^{\text{col}} + r_k^{\text{goal}} + r_k^{\text{emo}} \tag{6}$$

where r_k^{col} penalizes the collisions with all surrounding objects, r_k^{goal} rewards progress toward reaching the goal, and r_k^{emo} denotes an emotion-related penalty. Specifically, these three rewards are defined as follows.

1) Collision avoidance reward

$$r_k^{\text{col}} = \begin{cases} -0.6, & \text{if } d_{\min}^{\text{all}} \le r^{\text{robot}} \\ -0.1, & \text{else if } d_{\min}^{\text{static}} < d^{\text{static}} + r^{\text{robot}} \\ 0.0, & \text{else} \end{cases} \tag{7}$$

where $d_{\min}^{\text{all}}$ represents the minimum distance from the ego-agent to the margins of all surrounding objects as shown in Fig. 4, $d_{\min}^{\text{static}}$ denotes the minimum distance from the ego-agent to all static obstacles, d^{static} stands for a safety distance expanded from the collision margin of the obstacle, and r^{robot} is the ego-agent radius.

2) Goal reaching reward

$$r_k^{\text{goal}} = \begin{cases} 0.5 & \text{if } d_k^{\text{goal}} \le d^{\text{goal}} \\ 0.3 \cdot (d_{k-1}^{\text{goal}} - d_k^{\text{goal}}) & \text{else} \end{cases} \tag{8}$$

where d_k^{goal} stands for the distance from the ego-agent to the goal at the kth walking step, and d^{goal} represents the goal-reaching threshold. In particular, the second term encourages the robot to move toward the goal, thus alleviating the sparsity issue of the reward function and promoting efficient training.

TABLE I
PARAMETER (PARAM) SETUP

Param	Value	Param	Value	Param	Value
T	0.4 s	H	1.02 m	g	9.81 m/s^2
γ	0.99	d^{static}	0.2 m	$\bar{u}^v$	0.4 m/s
N	9	d^{goal}	0.1 m	$\bar{u}^{\Delta\theta}$	0.2 rd
r^{ped}	0.3 m	r^{static}	$[0.2, 0.4]$ m	r^{robot}	0.3 m

3) Emotion-aware reward

$$r_k^{\text{emo}} = \begin{cases} -0.1, & \text{if } \mathcal{C} \\ 0.0, & \text{else} \end{cases} \tag{9}$$

where $\mathcal{C}$ is a distance condition with respect to pedestrian emotions. Specifically, the discomfort distance of the ith pedestrian is defined as $d_{p_i}^{\text{emo}} \in \{0.2, 0.35, 0.5\}$m, depending on the corresponding emotion of the pedestrian {`happy`, `neutral`, `negative`}. r^{ped} is a constant pedestrian radius. Given the distance d_{p_i} from the margin of the ith pedestrian to the ego-agent, the penalty is -0.1 if $\exists\, i$, such that $d_{p_i} < d_{p_i}^{\text{emo}} + r^{\text{robot}}$. Note that, to avoid learning a conservative navigation policy, we only count once if the ego-agent enters the discomfort zones of several objects, including static obstacles and dynamic pedestrians, at the same walking step. That means we add -0.1 only once in (6) although it appears in both (7) and (9).

Remark 3: Identifying the optimal reward configuration is challenging due to the highly coupled relationships among different reward terms. For example, when larger penalty factors were applied for entering discomfort zones (e.g., -0.6), the number of discomfort events decreased; however, this came at the cost of increased navigation time, as the agent adopted more conservative behaviors. Due to space limitations, we have chosen not to include the detailed results of these parameter variations. Instead, we empirically fine-tuned the reward function and ultimately selected a feasible and suboptimal combination that balances safety and efficiency.

Our algorithm is summarized in Algorithm 1.

IV. EXPERIMENTS

We first detail the parameter specifications and the network architecture, followed by descriptions of the environmental settings. Subsequently, we experimentally evaluate the tracking errors introduced by the use of ROMs. Next, we benchmark our proposed navigation framework against SOTA methods that exclude emotion-related observations and reward functions. To further demonstrate the emotion awareness, we also implement a variant trained with emotion inputs and rewards. Further analyses investigate the influence of time-varying emotions. Finally, we present hardware deployments to demonstrate the real-world applicability of our approach.

A. Parameter Specifications and Network Details

The parameter specifications are shown in Table I. We set $\bar{u}^v = 0.4\,\text{m/s}$ and $\bar{u}^{\Delta\theta} = 0.2\,\text{rad}$, representing a reasonable linear velocity and turning angle to ensure stable locomotion. The geometry parameters r^{ped}, r^{static}, and r^{robot} are defined based

Algorithm 1: EmoBipedNav Algorithm.

Initialize: SAC replay buffer $\mathcal{D}$, $k \leftarrow 1$
Reset: pedestrian positions and emotions, positions and sizes of static obstacles, robot position, and $\mathbf{u}_{k-1} \leftarrow \mathbf{0}$
Observe: $\mathbf{o}_k^{\text{sp}-\text{tem}}$, $\mathbf{s}_k^{\text{goal}}$, and $\mathbf{u}_{k-1}$
while $k \leq \mathcal{K}$ **do**
    Sample action $\mathbf{u}_k$ using (5a) and (5c)
    Compute the desired foot placement with ALIP [32] given $u_k^v \in \mathbf{u}_k$
    Track the full-body trajectory planned based on the foot placement using a passivity-based controller [36]
    Rotate hip yaw-joint to track heading angle $u_k^{\Delta\theta} \in \mathbf{u}_k$
    Joint torque control for Digit
    $k \leftarrow k + 1$
    Observe
    Calculate reward with (6)–(9)
    if goal reached **or** collision **or** timeout **then**
        Reset
    end if
    Add observation, reward, and reset information to $\mathcal{D}$
    Update networks (5) using samples from $\mathcal{D}$ and SAC Algorithm
end while

on the approximate sizes of the corresponding objects, including pedestrians, static obstacles such as chairs and trash cans, and the ego-agent. The LIP model parameters, T and H, are selected according to the kinematics configuration of the Digit robot. The discount factor $\gamma = 0.99$ is a common choice for DRL problems. The parameter $N = 9$ is empirically determined, where a larger N allows for capturing more historical information but results in computationally heavier neural networks.

Our DRL network architecture consists of three modules: encoder, actor, and critic. The encoder comprises four layers of CNNs, each with 32 kernels of size 3. The stride of the first layer is 2, while the remaining layers have a stride of 1. After the CNN layers, a fully connected layer extracts 50 features from the sequential LiDAR maps. Both the actor and critic networks use similar MLP architectures, each with two hidden layers of size 1024. Unlike critic networks, the output of the network has only one value, while the network of actors produces 4 variables, corresponding to 2 means and 2 $\log$ standard deviations for the action $\mathbf{u}_k$. The learning rates for both the actor and critic networks are set to 0.001.

B. Environment Settings

The simulation environment includes both static obstacles and dynamically moving pedestrians interacting with a full-body bipedal robot.

Static obstacles: Static obstacles are randomly placed within a rectangular area, with the x-coordinate ranging from -3 to 3m and the y-coordinate from -1.5 to 1.5 m. Each obstacle has a radius ranging from 0.2 to 0.4 m.

Pedestrians: The initial position of each pedestrian is distributed along the perimeter of a circle with a radius randomized from 3.5 to 4.5 m. The corresponding goal of each pedestrian is located on the other side of the circle. Pedestrians move between these two points at a preferred speed of 1 m/s.

Pedestrian motion is generated using the optimal reciprocal collision avoidance (ORCA) method [35], a multiagent motion planner commonly used to simulate interactive motions of all agents [23], [24], [25], [27], [29]. In addition to pedestrian–pedestrian interactions, we also enable pedestrian–robot interactions when the pedestrian's speed exceeds the robot's speed. Otherwise, the robot remains invisible to pedestrians, and only the robot actively avoids collisions with pedestrians.

Robot: The robot navigates from position $(-4, 0)$ to position $(4, 0)$ m. We simulate the bipedal robot Digit in MuJoCo [34]. The navigation policy in the high-level motion planner only generates the sagittal speed and heading angle derived from the ROM.

To track the desired sagittal speed, we first employ the ALIP planner [32] to compute the swing foot placement. Based on this placement, we define the swing foot trajectory, which is then used to plan the full-body joint motions via inverse kinematics. To address gaps in robot dynamics, we incorporate a passivity-based controller [36], which effectively preserves the natural dynamics of the system while ensuring asymptotic tracking performance. In our previous work [16], [21], the addition of toe–joint actuation alongside the whole-body passivity controller demonstrated promising results in minimizing the mismatch between the LIP model trajectory and the full-order trajectory of the Digit bipedal robot.

C. Model Discrepancies

We describe the model discrepancies between the ROM and full-body bipedal dynamics in Section III-A. To clearly and intuitively illustrate these model gaps, we conduct simulation experiments that evaluate the tracking performance.

We select a representative simulation trial in MuJoCo, where Digit navigates a pedestrian-rich environment. For each walking step, we plot the desired x and y positions and the heading angle, alongside the actual positions and heading angle of the simulated Digit. The tracking results are shown in Fig. 5. While the tracking of the x-position is highly accurate, the y-position tracking error is nonnegligible, and the heading angle exhibits more pronounced deviations. As discussed in (1), the ROM is primarily designed for sagittal motion and does not explicitly account for body orientation and lateral motion. As a result, x-direction tracking remains precise, whereas tracking errors in y-position and heading angle become more noticeable, particularly when the desired heading angle changes frequently. Such tracking errors significantly degrade navigation performance when deploying policies trained with ROMs onto full-body bipedal robots, as will be experimentally demonstrated in Section IV-E.

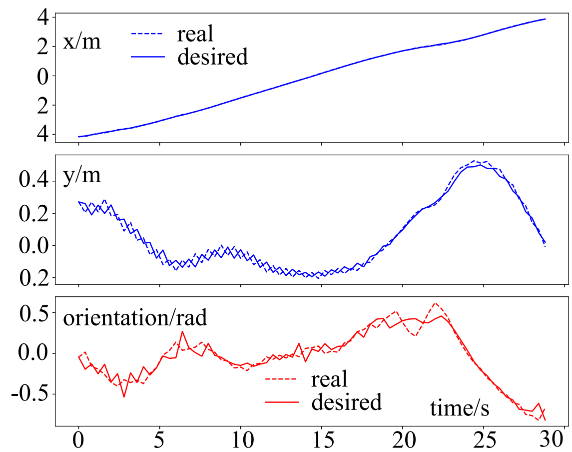
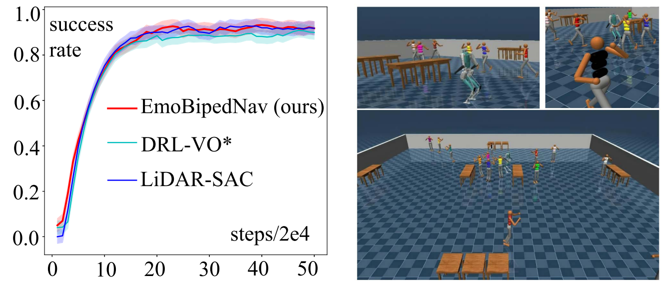


Fig. 5. Tracking performance. The top figure shows the x tracking, the middle plots the y tracking, and the bottom illustrates the heading angle following in the world frame.



(a) (b)

Fig. 6. Training process with simulated Digit in MuJoCo. (a) Training process. (b) Simulation scene in MuJoCo. We evaluate each model 100 times at every 2×10^4 training steps and compute corresponding success rate. (a) Training process. (b) Training scene in Mujoco

TABLE II
FINAL PERFORMANCE EVALUATION ON THE FULL-BODY BIPEDAL ROBOT WITH 500 RANDOM TESTS

Method	SR	NT	DT
DWA	0.892	**27.20**±3.55	415
LNDNL	0.660	38.13±5.99	301
DRL–VO*	0.912	28.16±**2.49**	318
LiDAR–SAC	0.932	27.24±3.10	766
Single–LGM	0.928	33.15±4.84	546
OGM	0.944	29.45±3.79	312
EmoBipedNav (ours)	**0.950**	28.83±3.15	**226**

Bold values indicates the best performance.

D. Benchmarks

To highlight the superior social navigation capabilities of our approach against existing methods, we first disable emotion-related observations and reward functions, ensuring a fair comparison with SOTA baselines. The SOTA baselines are listed below.

DWA: The dynamic window approach (DWA) [37] is a traditional nonlearning method widely applied for local navigation using wheeled mobile robots.

LNDNL: The LNDNL method [28] is a DRL-based navigation approach that directly maps the raw LiDAR scans to robot actions.

LiDAR–SAC: LiDAR–SAC is inspired by the SOTA study in [27], using an encoder to extract latent features from sequential LiDAR scans stacked over the time horizon. We replace the original networks with ours due to learning failures.

DRL–VO: The DRL–VO pipeline [29] combines fully observable states, such as pedestrian positions and velocities, with observations of raw sequential LiDAR scans. We improve the pipeline by integrating our SAC structure, resulting in the modified DRL–VO*.

SARL and RGL: SARL [24] and RGL pipelines [25] assume fully observable environments. However, when the ego-agent is a bipedal robot with velocity constraints in (2), neither SARL nor RGL succeed in achieving a feasible navigation policy.

Single-LGM: Single-LGM uses only one single LiDAR grid map (LGM) at the current walking step instead of a sequence of historical ones, as illustrated in Fig. 2(a).

OGM: OGM leverages a sequence of historical OGMs shown in Fig. 3(c) in the frame of the ego-agent, replacing the LGMs illustrated in Fig. 3(b).

Evaluation metrics: We evaluate each benchmark method over 500 random trials. The initial positions of dynamic and static objects are set randomly for each evaluation, while the initial and goal positions of Digit's torso remain the same. For each trial, all benchmarks start with the same initial environments, ensuring that dynamic and static objects have the same initial positions and pedestrian emotions are consistent. The success rate (SR) is defined as the ratio of successful episodes that occur without collision or timeout. The navigation time (NT) represents the average time taken for all successful episodes. Discomfort times (DT) count the number of walking steps when Digit enters the discomfort zone of a static obstacle or a pedestrian.

As illustrated in Fig. 6, both LiDAR–SAC and DRL–VO* produce results similar to ours when their original network structures are replaced with the architecture described in Section III, further validating the robustness and effectiveness of our network pipeline. Moreover, DRL-VO* has the same network structure as ours but yields inferior performance in terms of the stable success rate, further indicating that the proposed LGMs are better representations. In addition, LNDNL is omitted from Fig. 6 because its success rate climbs to 0.66 and decreases to 0. Furthermore, SARL and RGL are excluded from the analysis as they fail. The trainings of Single–LGM and OGM are removed due to their similar learning curves to our method.

The quantitative results, presented in Table II, confirm that our method achieves the highest SR for the deployment on the full-body bipedal robot Digit. DWA enables efficient navigation with the shortest average time of 27.2 s, at the cost of decreasing SR (0.892) and increasing DT. It is noteworthy that DWA can reach a higher SR (0.950) when deployed with the LIP model without considering full-order dynamics, which underscores the notable dynamics discrepancies between the ROM and the full-body robot dynamics. LNDNL demonstrates significantly inferior performance in terms of both SR and

TABLE III
PERFORMANCE EVALUATION W/O EMOTIONS AND W/O FULL-BODY ROBOT DYNAMICS WITH 500 RANDOM TESTS

DD (Train)	Train/Test	Obs	IDT (Test)	SR
0.2 m	Digit/Digit	LGM	[**51**; 50, 181, 1352]	0.950
0.5 m	Digit/Digit	LGM	[395; 81, **138**, **1167**]	0.938
Var-DD	Digit/Digit	LGM	[213; 56, 166, 1198]	0.940
Var-DD	Digit/Digit	OGM	[436; 87, 160, 1174]	0.892
Var-DD	LIP/LIP	LGM	[202; **37**, 143, 1227]	**0.962**
Var-DD	LIP/Digit	LGM	[642; 66, 177, 1266]	0.900

DD: discomfort distance (m); Obs: observation
Var-DD= $[0.2, 0.35, 0.5]$ m $\leftrightarrow$ [happy, neutral, negative]

Bold values indicates the best performance.

NT due to the plateauing problem during training. Notably, we observed that LNDNL achieves results comparable to our approach when trained and evaluated with the ROM without incorporating full-body dynamics, emphasizing our pipeline's capability to effectively handle the complex dynamics of the full-body robot. Although single–LGM can also achieve a high SR, its NT significantly increases due to its conservative policy, which does not account for the temporal dynamics of the social environment. OGM, on the other hand, shows a comparable SR, but its DT is higher than our method, indicating that LGMs are more effective in reducing discomfort.

E. Emotion-Integrated Implementation

To investigate further, we conduct an additional training session to assess the performance of integrating emotion observations. The results demonstrate that our approach can adapt to pedestrians with different emotions, showing the emotion-aware feature of our navigation policy. In addition, we validate the performance discrepancies of social navigation when using the ROM and the full-body bipedal robot. Moreover, we practically verify that using LGMs provides greater benefits for collision and discomfort avoidance compared to using OGMs in these emotion-aware environments.

First, we use variable discomfort distances (Var–DD) indicating different emotions of pedestrians. Next, we set the discomfort distance to a constant 0.5 m. Furthermore, we replace LGMs with OGMs. Finally, we train our pipeline using only the LIP model.

To account for pedestrian emotions and their associated discomfort distances, we introduce an additional evaluation metric, IDT, which quantifies the discomfort times for individual entities. Specifically, the four values in the IDT column of Table III correspond to the IDT of static obstacles and pedestrians exhibiting happy, neutral, and negative emotions, respectively. During training, the two baseline models enforce a fixed discomfort distance of 0.2 or 0.5 m for all dynamic pedestrians, regardless of their emotional states. However, during testing, IDT is computed based on variable discomfort distances of $[0.2, 0.35, 0.5]$ m, corresponding to the three emotional states. This discrepancy between training and testing conditions allows us to evaluate the model's ability to adapt to varying discomfort zones.

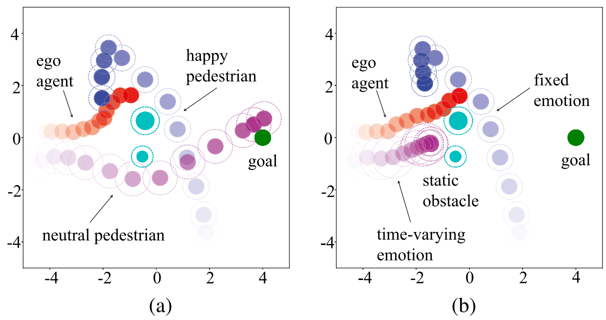


Fig. 7. Robot and pedestrian trajectories under different emotion settings. (a) Fixed emotions. (b) Time-variant emotions. The decreasing transparency illustrates the time evolvement. (a) Fixed emotion. (b) Time-varying emotion.

Compared to maintaining a constant discomfort distance of 0.2 m, our approach of using variable discomfort distances of $[0.2, 0.35, 0.5]$ m decreases IDT for neutral and negative pedestrians while increasing IDT for happy pedestrians, and static obstacles in particular. This indicates that our approach prioritizes neutral and negative pedestrians by assigning them longer discomfort distances. However, this comes at the cost of reduced comfort for happy pedestrians and, notably, static obstacles. The performance degradation in the presence of static obstacles may stem from reduced available free space. This reduction arises because the discomfort zones associated with neutral and negative pedestrians can expand significantly, while the overall environment remains unchanged. Moreover, the ego-agent is inherently more susceptible to collisions with moving pedestrians due to their dynamic nature. As a result, it prioritizes maintaining distance from them, which can inadvertently increase the likelihood of intruding into the discomfort zones of static obstacles.

When the constant discomfort distance is increased to 0.5 m, the ego-agent further improves IDT for neutral and negative pedestrians, yet compromising comfort for happy pedestrians and static obstacles. The increased IDT for static obstacles could be attributed to the same factors observed when using variable discomfort distances $[0.2, 0.35, 0.5]$ compared to a fixed discomfort distance of 0.2 m. In addition, the larger discomfort distance leads to a decline in SR, which may stem from the aggressive pedestrian motion generated by ORCA [35], despite the conservative behaviors of the ego-agent for discomfort and collision avoidance.

When replacing our LGMs with OGMs, both SR and IDT of static obstacles significantly degrade, further demonstrating the effectiveness of LGM in enhancing collision and discomfort avoidance. In addition, while the pipeline trained and tested with the LIP model achieves the highest overall performance, its SR drops to 0.9, and IDT increases substantially when tested with Digit, further highlighting the limitations of ROMs in maintaining robust navigation performance.

F. Implementation With Time-Varying Emotions

In real-world scenarios, emotions can fluctuate or be misclassified due to inaccuracy in the facial recognition algorithm,

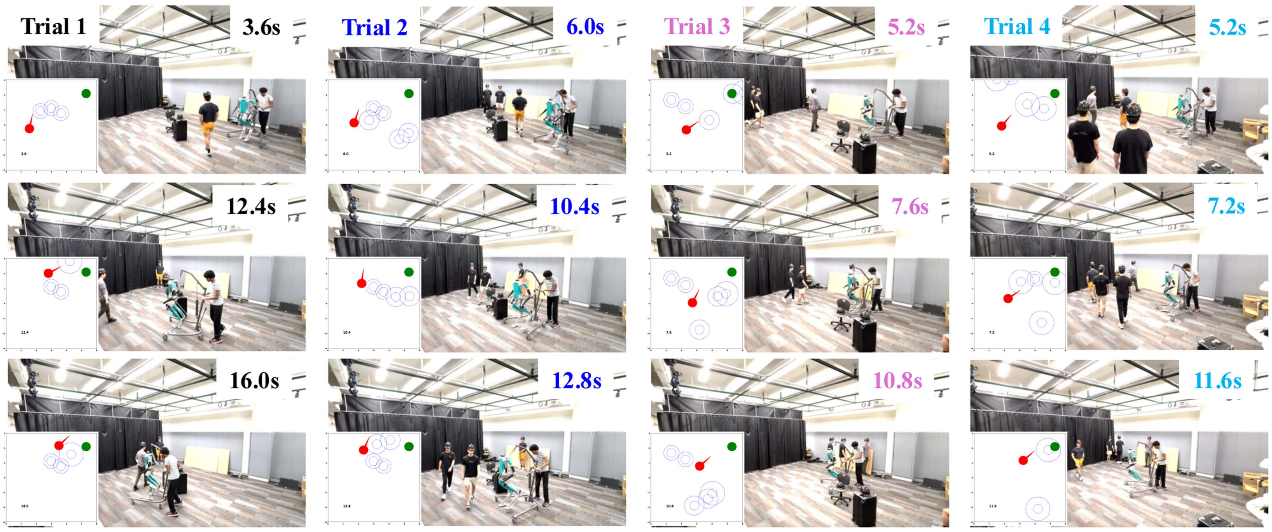


Fig. 8. Screenshots of hardware deployments under various pedestrian navigation patterns. Each column corresponds to an implementation. In each screenshot, a LiDAR scan figure is overlaid to illustrate the robot's position and orientation, represented by a solid red circle and a red arrow, respectively. Object collision margins are depicted as blue hollow circles, while discomfort margins are indicated with dashed circles. We track the robot and pedestrians with an external motion capture system for accurate trajectory analysis. Pedestrian emotions are randomly set at the beginning and remain constant for each implementation.

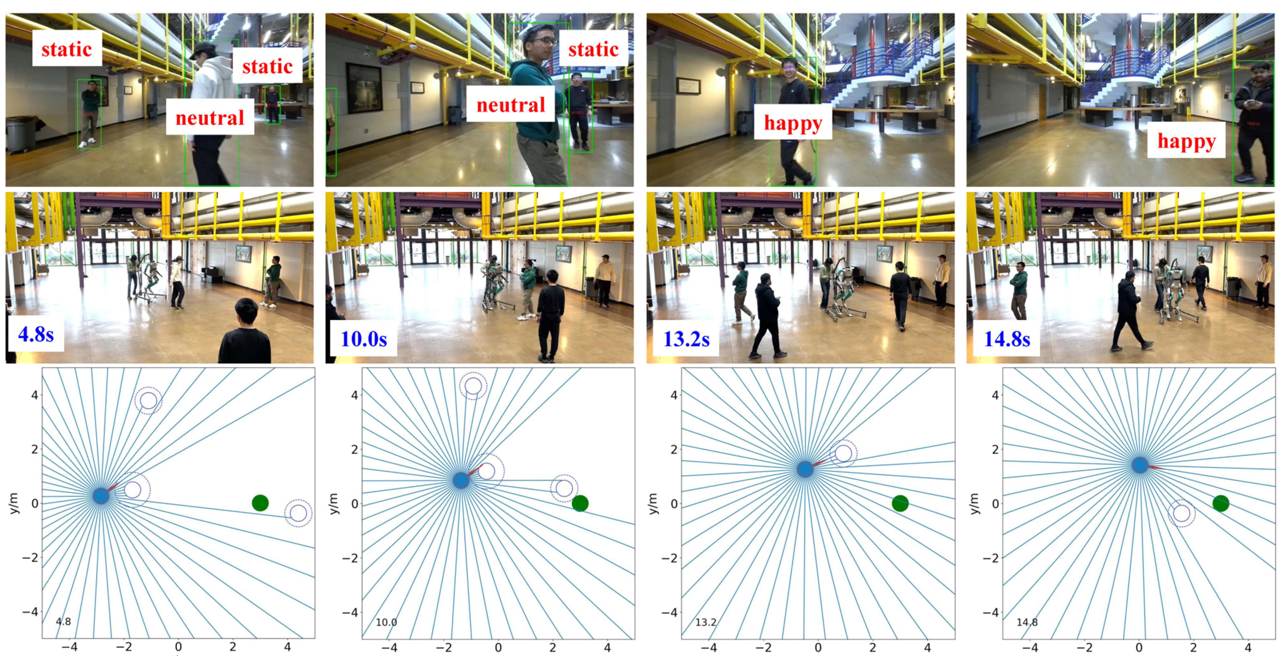


Fig. 9. Emotion recognition with an on-board ZED camera. The top row presents the ZED camera's perspective, showcasing detected pedestrians along with their recognized emotions. The middle row shows the recorded scenes from a fixed external camera view. The bottom figures illustrate a top-down view of the reconstructed LiDAR scan, position, and orientation of the ego-agent, and collision and discomfort margins of pedestrians.

which is a pretrained CNN model relying on small dataset.[1] To investigate the adaptability of our approach to these uncertainties, we introduce a more challenging simulation setup where one pedestrian's emotion keeps alternating among `happy`, `neutral`, and `negative` every 2 s. Statistically, `SR` roughly remains the same, while `IDT` shifts from $[213; 56, 166, 1198]$ to $[258; 42, 204, 1317]$. The similar `SR` shows that our approach is stable for collision avoidance, although pedestrian emotions vary over time. The increased `IDT` for `neutral` and `negative` pedestrians may result from the ego-agent intruding into larger discomfort zones when nearby pedestrians suddenly change to a `negative` emotional state from `neutral`, or from `happy` to `neutral`. Fig. 7 presents a representative episode, comparing the case with fixed emotions

[Fig. 7(a)] to that with time-varying emotions [Fig. 7(b)]. The latter scenario shows increased discomfort times, particularly at the beginning of the episode when a nearby pedestrian frequently changes the emotion, particularly from `neutral` to `negative`.

G. Hardware Demonstrations

Sim-to-real approaches: Domain randomization is a commonly used technique to facilitate sim-to-real transfer. In the simulation settings, as described in Section IV-B, we randomize the number, emotions, and initial positions of pedestrians, as well as the number, sizes, and positions of obstacles. In addition, enhancing simulation fidelity further benefits sim-to-real transfer. We directly train our navigation policy using a simulated Digit robot, rather than a point-mass model or a ROM, thereby improving its transferability from simulation to real-world deployment. Furthermore, in real-world environments, we first localize surrounding objects in the ego-agent frame and approximate each as a circle with a fixed radius of 0.3 m for simplification. Then, we emit K rays from the ego-agent to intersect these circular approximations, generating a reconstructed LiDAR scan that more closely replicates the structure of simulated observations.

Adaptability for sim-to-real deployment: Although it is challenging to quantitatively assess how capable the trained policy is to adapt and generalize when deployed in previously unseen real-world environments, its generalizability can still be evaluated in simulation through unexplored scenarios. In our training environments, the maximum number of pedestrians was four and the maximum number of static obstacles was three. In one test scenario, we chose the number of pedestrians at six and varied the number of static obstacles from one to three. In another scenario, we removed moving pedestrians and fixed the number of static obstacles at six. We conducted 500 random trials and obtained success rates of 0.802 and 0.898 for the two test scenarios, respectively. Although our policy was trained in relatively simple environments, it remained transferable to more complex scenarios, albeit with degraded performance due to the increased environmental complexity.

Deployments under various pedestrian motion patterns: We begin by transferring the simulated policy learned in MuJoCo [34] to the physical robot Digit. Furthermore, we include various pedestrian motion patterns to verify the sim-to-real capabilities of our emotion-aware navigation policy. Fig. 8 shows four real-world implementations. The first three columns depict scenarios involving dynamic pedestrians and static obstacles, while the last row showcases an environment where all pedestrians are in dynamic movement. Notably, in the first column, a pedestrian intentionally walks in front of Digit and remains stationary to obstruct its path. In addition, pedestrians move ingroup together in the second and third columns. Despite these diverse and challenging conditions, Digit successfully reaches its goal without collision while actively avoiding intrusions into discomfort zones.

Deployments integrating emotion recognition: To validate our pipeline with integrated pedestrian emotion recognition, we deploy it on the Digit robot equipped with a ZED stereo camera. The robot is localized using LiDAR SLAM. Pedestrians are detected at 10 Hz on a Jetson Orin, while face extraction and emotion recognition run at 0.5 Hz using ZED and OpenCV APIs. An Intel NUC handles SLAM at 10 Hz, the low-level controller at 1000 Hz, and the social navigation policy at 2.5 Hz.

Real-world deployments, as demonstrated in the accompanying video, confirm the seamless transferability of our navigation framework from simulation to real-world applications. Fig. 9 provides representative screenshots from a hardware demonstration. When pedestrians are far from Digit or facing away from the camera, their faces cannot be detected and are therefore labeled with `neutral` emotions. Once their faces become clearly visible, their emotions change to `happy`. In the accompanying video, we showcase additional scenarios where pedestrian emotions transition from `neutral` to `negative` or shift between other emotional states. Furthermore, we operate Digit on campus with more realistic social interactions, further demonstrating the practicality of our emotion-aware social navigation pipeline for bipedal robots (see the attached video).

V. Discussion

In this study, we present a DRL-based navigation framework designed to enable bipedal robots to navigate socially interactive environments integrated with pedestrian emotions. Future research will focus on incorporating more realistic social norms, such as adhering to conventions like "walking on the right," and conducting DRL training in increasingly complex social scenarios, including walls and grouped pedestrians. The ultimate goal is to facilitate natural and socially compliant locomotion for bipedal robots in crowded environments.

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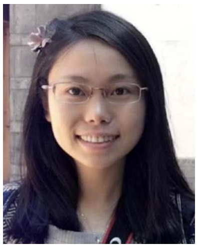

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